

High Myopia and Management

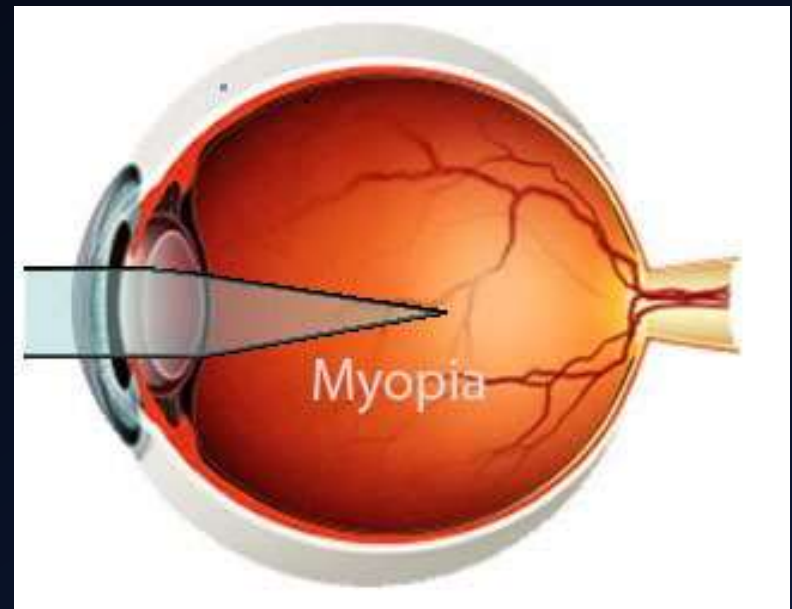
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16th Batch

Presentation Layout

- ✓ Introduction to myopia and its classifications
- ✓ High Myopia – Etiologies
 - Symptoms and signs
 - Complications
- ✓ Types of high myopic correction
 - Spectacles
 - Contact lenses
 - Refractive surgeries
- ✓ Low vision management in high myopia

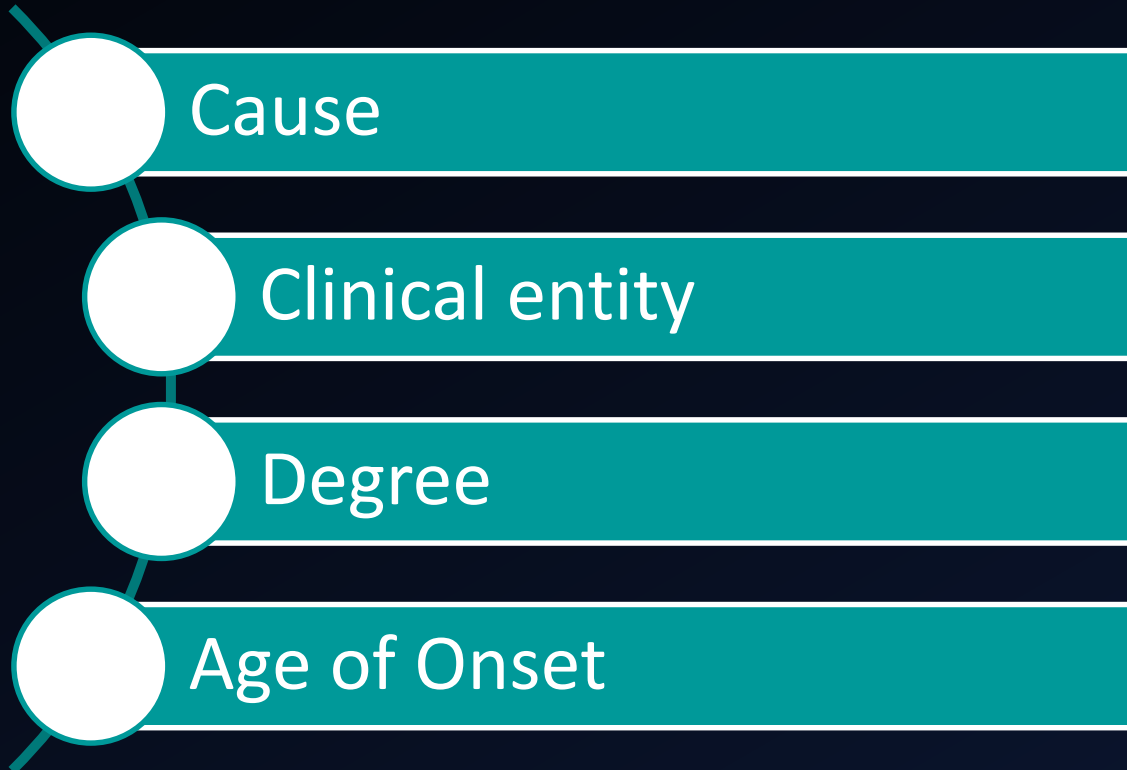
Myopia

- Derived from Greek word “muopia” – to close the eyes
- Refractive anomaly of the eye in which light rays from an object at infinity entering a non accommodating eye are focused in front of the retina



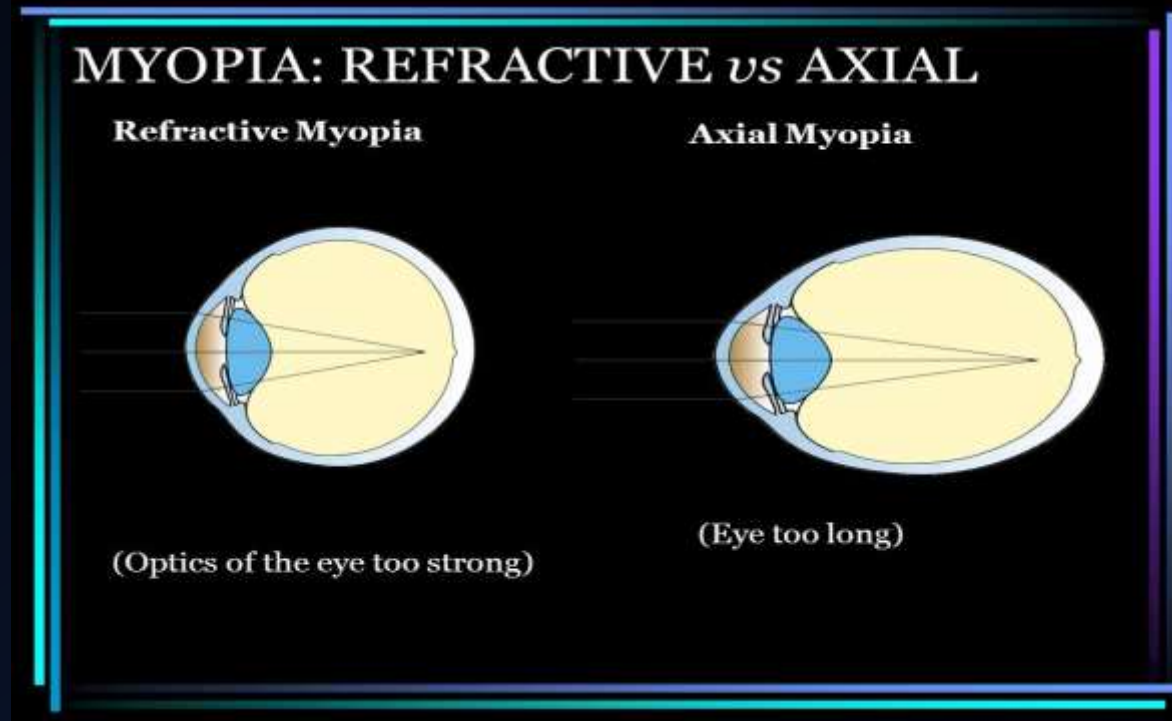
Classification Systems for Myopia

- can be classified on the basis of:



Cause

1. Axial: eye too long for its refractive power
2. Refractive: refractive system too powerful for the axial length of the eye



Clinical Entity

1. Simple Myopia
2. Nocturnal Myopia
3. Pseudomyopia
4. Degenerative Myopia
5. Induced Myopia

Age of Onset

1. Congenital Myopia (present at birth and persisting through infancy)
2. Youth Onset Myopia (< 20 years of age)
3. Early Adult Onset Myopia (20 – 40 years of age)
4. Late Adult Onset Myopia (> 40 years of age)

Degree

1. Low Myopia (< 3.00 D)
2. Medium Myopia (3.00 D – 6.00 D)
3. **High Myopia (> 6.00 D)**

HIGH MYOPIA

- Spherical equivalent exceeding -6.00 D and/or the axial length longer than 26.5 mm
- High myopia with any posterior myopia specific pathology resulting from excess axial elongation



Degenerative/Pathological Myopia

Etiologies

- **Role of heredity**

- High myopia considered to be a sex linked, recessive inherited disorder
- Heredity linked growth of retina is the determinant in development of myopia

- **Role of general growth process**

- Minor role
- Lengthening of the posterior segment of globe commences only during the period of active growth and ends with the termination of active growth

Genetic factors
(play major role)

General growth process
(play minor role)

More growth of retina

Stretching of sclera

Increased axial length

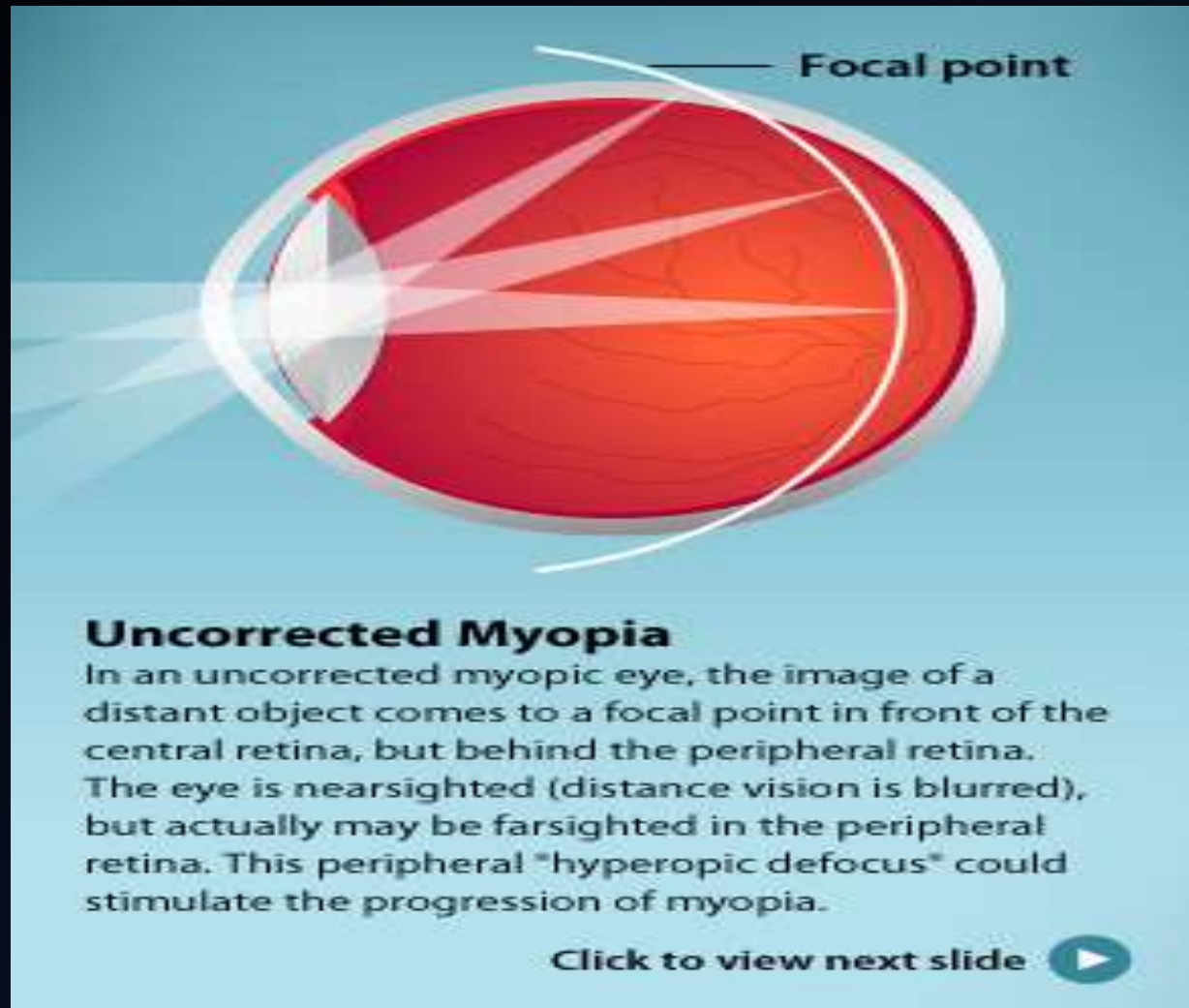
Degeneration of choroid

Degeneration of retina

Degeneration of vitreous



Peripheral hyperopic defocus



Risk factors for High Myopia

- Inheritance
- Premature birth
- Systemic associations
 - Down syndrome
 - Marfan syndrome
 - Stickler syndrome
 - Pierre – robin syndrome

Symptoms of High Myopia

1) Defective vision

- 
- Due to uncorrected high errors
 - Uncorrective loss of vision due to progressive degenerative changes

2) Closer working distance for near tasks

e.g . -10.00 D myope, working distance 10 cm or less

3) Delayed dark adaptation

In high myopes with healthy fundus:

Dark adaptation prolonged with normal retinal sensitivity

In high myopes with retinal degeneration:

Dark adaptation prolonged with reduced retinal sensitivity

Correlation between dark adaptation and refractive errors; Indian Journal of Ophthalmology 1985

4) Night Blindness

- In very high myopes having marked chorioretinal degeneration

5) Recent onset or increase in floaters

6) Flashing light

7) Curtain effect coming down across vision

PVD / Retinal Detachment



Peripheral vision in high myopia

Part of vision that occurs outside the very center of gaze

- Part of vision that occurs outside the very center of gaze
- From 1 to 2 mm retinal eccentricity
- Cone density decreased from 18,500/mm² to 14,000/mm² in emmetropic eye
 - Cone density decreased from 9,100/mm² to 8,200/mm² in myopic eye

(Effect of myopia on cone photoreceptor density , *Investigative Ophthalmology and Visual Science* 2007)

Accommodation in high myopia

- Accommodation need not develop normally resulting in
 - convergence insufficiency
 - exophoria

Color vision in high myopia

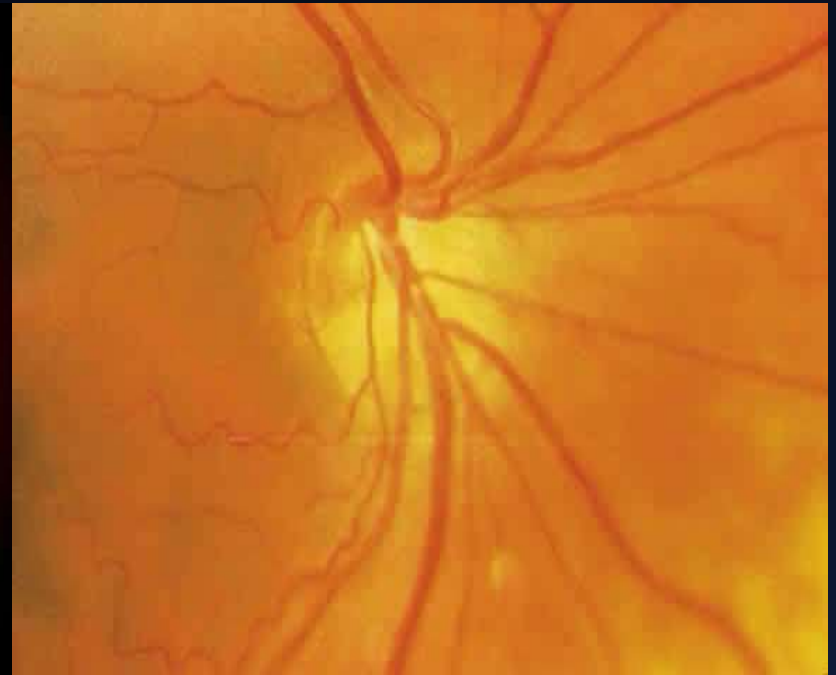
- In spectral sensitivity test, the red and green sensitivities of high myopes without degenerative fundus changes almost normal
- Sensitivity to the blue color was reduced
- S cones comprise only 10% of human retinal cones, more vulnerable to damage

(Color vision and dark adaptation in high myopia without central retinal degeneration, *British Journal of Ophthalmology* 1995)

Signs in high myopia

- Prominent eyes
- Large corneas
- Deep anterior chamber
- Slightly large pupil

Fundus Changes In High Myopia



Optic Disc Crescent

- Early change in the myopic fundus
- Is due to pulling away of choroid and pigment epithelium, usually from temporal edge of nerve to expose the sclera



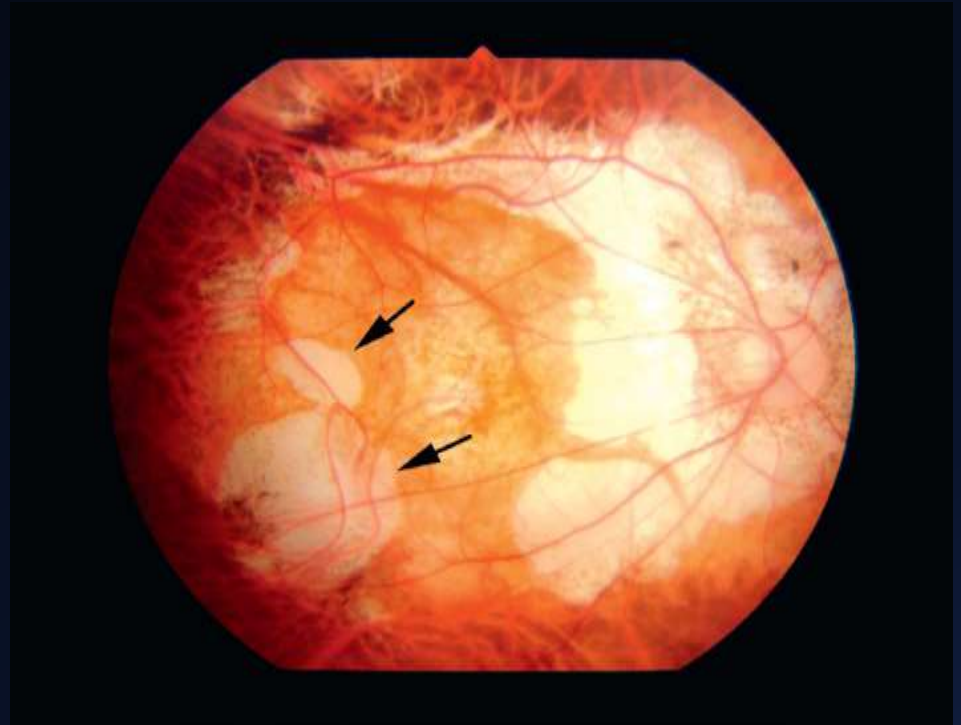
Tessellated fundus

- Hypoplasia of RPE following axial elongation reduces the pigment, allowing the visibility of choroidal vessels
- Aka diffuse chorioretinal atrophy



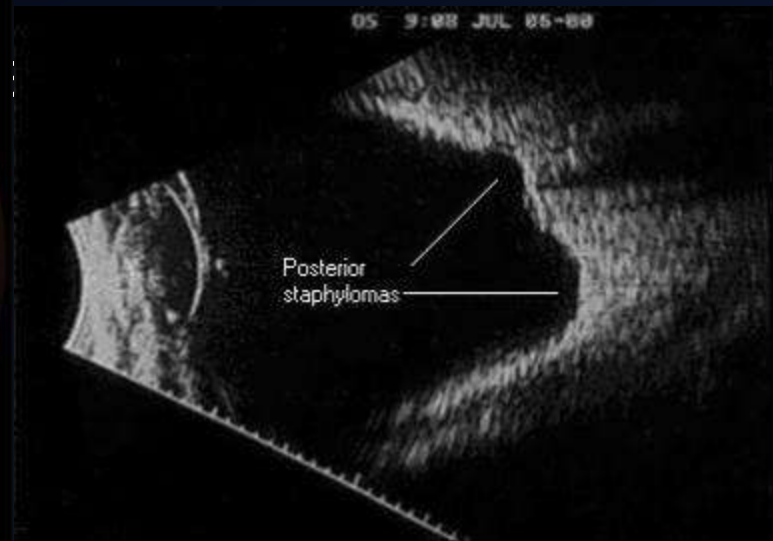
Patchy chorioretinal atrophy

- Grayish white and ill defined atrophy
- Caused by complete loss of choriocapillaries
- Corresponding absolute scotoma



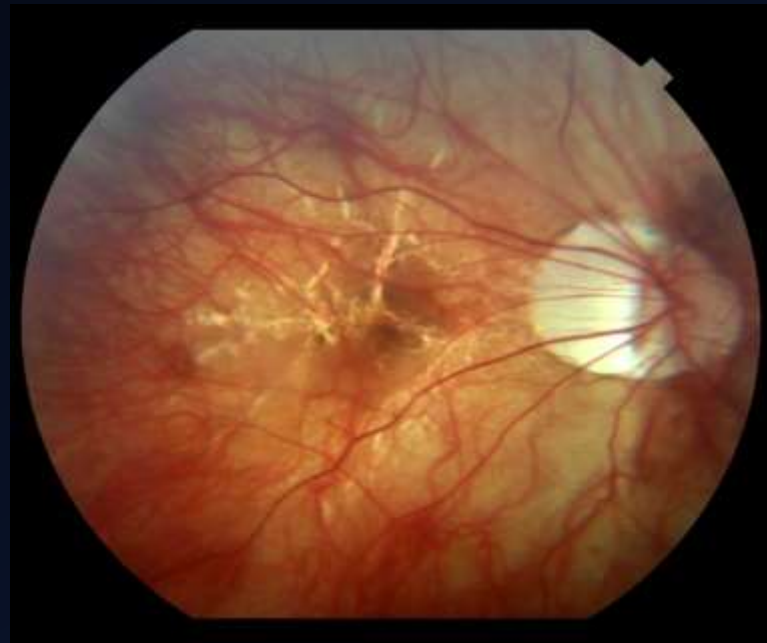
Posterior staphyloma

- Backward ectasia of the fundus
- Tesellation and pallor of the involved area
- More common in region of optic disc and macula
- Incidence higher in older patients (> 50 yrs) than in younger



Lacquer cracks

- Linear ruptures of Bruch's m/m in macular areas of highly myopic eye
- Observed as yellowish linear lesions in macula
- Present in about 4% of highly myopic eyes



Myopic choroidal neovascularisation

- Myopic CNV reduce the central vision
- Develops in 10% of highly myopic patients
- Myopic CNV not intensely active
- Progresses from an active phase to scar phase
- In scar phase, CNV is covered by proliferated RPE cells and is observed as dark pigmented spot



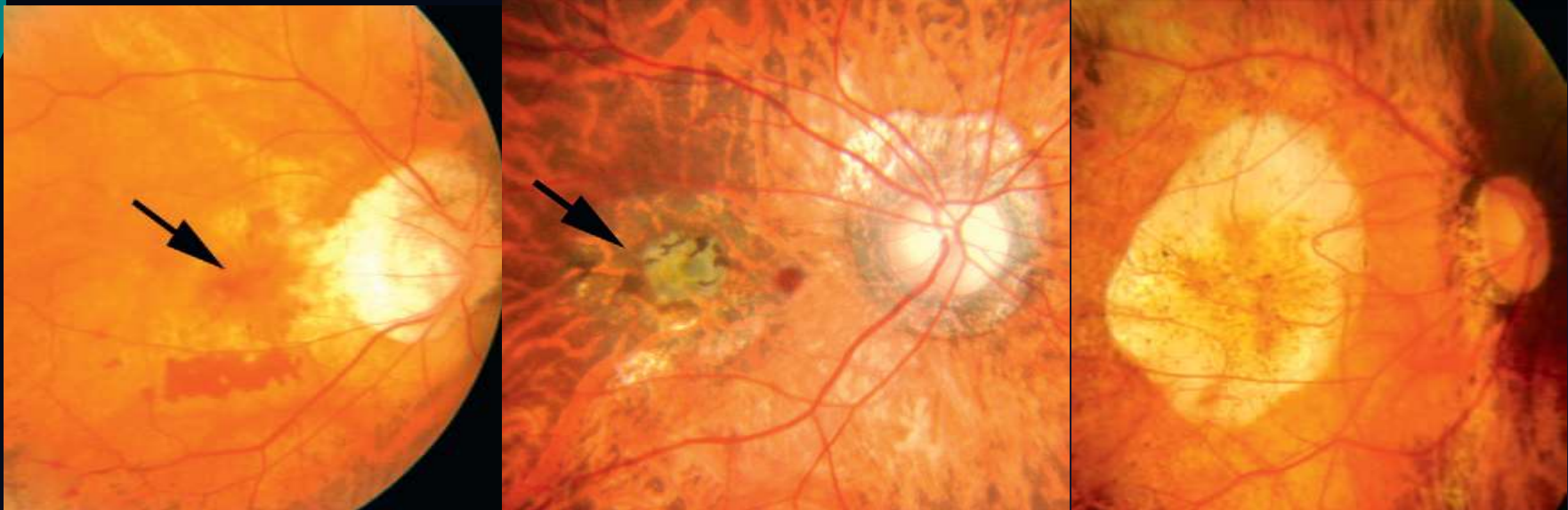
Fuchs spot

- After CNV regression, well defined chorioretinal atrophy develops and surround the Fuchs spot



Atrophic CNV

Progressive vision decrease in long term



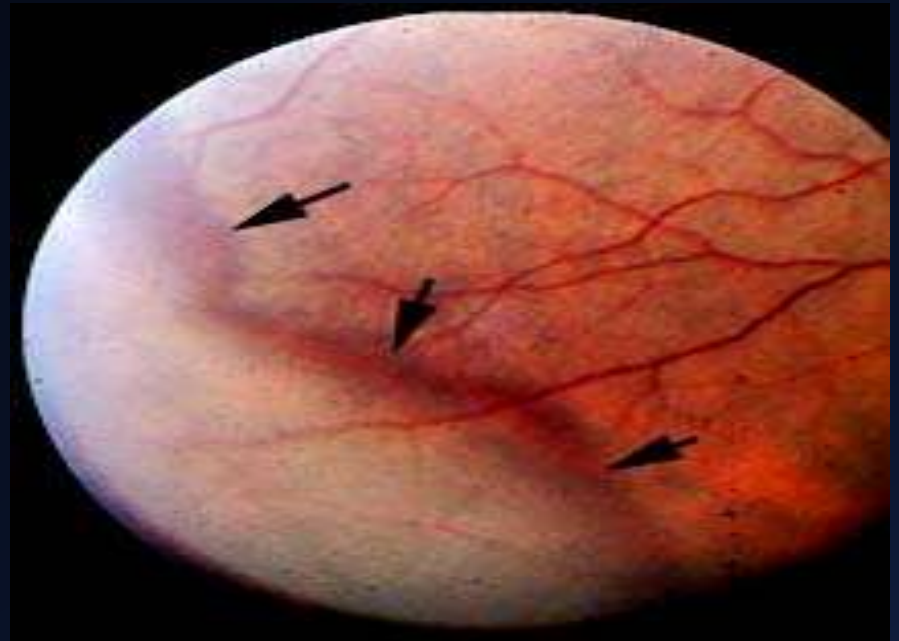
Peripheral retinal degenerations

- **Lattice degeneration**
 - Found more commonly in moderate myopes
 - Spindle shaped areas of retinal thinning
 - Located in temporal rather than nasal, superior rather than inferior fundus
 - Criss crossing thin white sc characteristic of latticewor



White without pressure

- Appears as whitish areas in peripheral retina
- Occurs without scleral indentation
- May indicate increased vitreoretinal adhesion and traction



Complications of high myopia

- Rhegmatogenous retinal detachment
- Macular hole
- Cataract
- Glaucoma
- Amblyopia



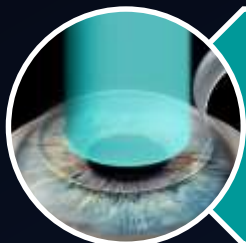
Types of High Myopic Correction



Spectacles



Contact lenses



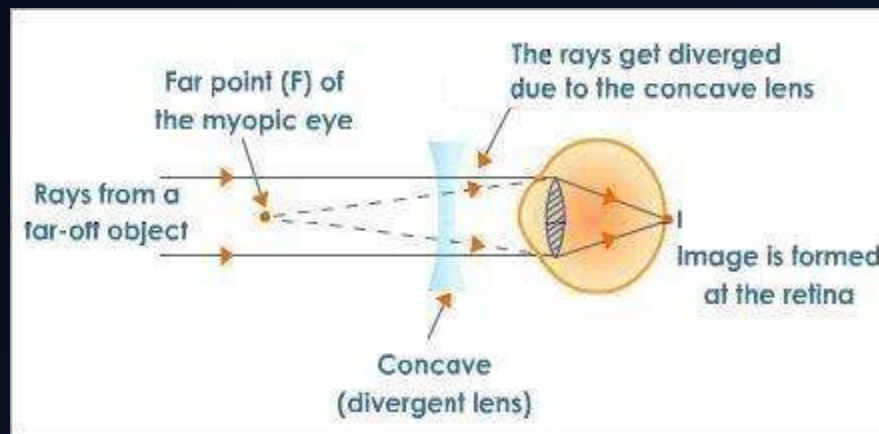
Refractive surgery

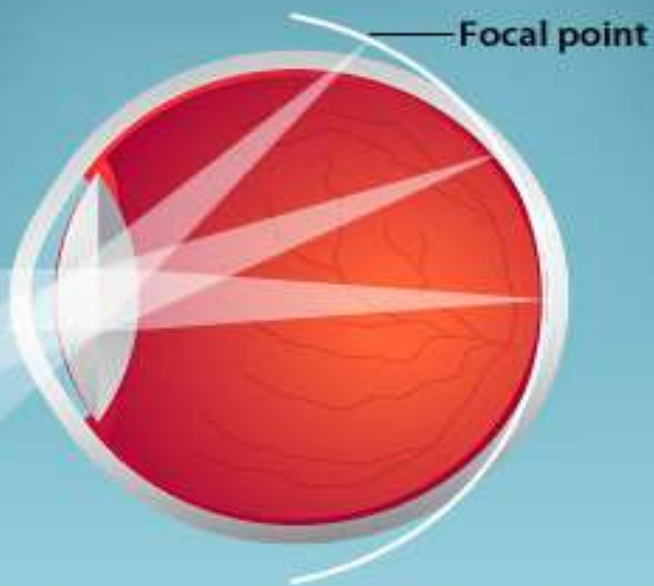
SPECTACLES

- A minus correcting lens whose secondary focal point coincide with the far point of the eye is placed at the spectacle plane



Artificially places the far point at infinity

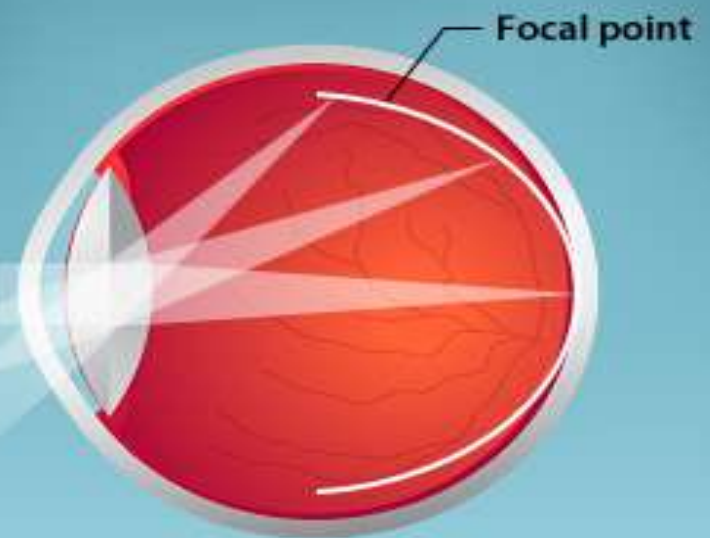




Traditional Myopia Correction

Traditional eyeglasses and contact lenses correct myopia at the center of the retina (making distance vision clear), but do not correct the relative hyperopia that exists in the peripheral retina. In fact, they might actually worsen the hyperopic defocus in the peripheral retina, which may increase the risk of myopia progression.

Click to view next slide



Optimal Myopia Correction?

Experimental "dual focus" contact lenses fully correct myopia and have a modified power in the lens periphery. This brings the peripheral focal point of the eye forward, replacing hyperopic defocus with myopic defocus (the focal point is now in front of the peripheral retina, not behind it), which might reduce the risk of increasing myopia.

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Advantages of Spectacle Correction



Easy



Safe



Inexpensive

Problems Caused by High Minus Lenses

- Lens weight and thickness
- Minification
- Field of view
- Edge reflection & concentric rings

1. Lens Weight and Thickness

- Lens weight is less of a problem for high myopes than for the aphakic
- Lens thickness more of a problem for high myopes than for aphakic



Thick edges much more obvious to the observer than thick centres

Minimizing lens weight

- **Plastic lenses**
 - Low specific gravity resulting in a lens of lighter weight

Material	Specific gravity	Index
Crown glass	2.54	1.523
CR-39 plastic	1.32	1.498
Polycarbonate	1.20	1.586

- **Small lens size**



Minimizing edge thickness

- Relationship between edge thickness and center thickness of lens:

$$t_c - t_p = F_A h^2 / 2(n-1)$$

t_c : center thickness

t_p : peripheral thickness

F_A : approximate power of the lens

h : half of lens diameter

n : refractive index



For a lens of a given power, edge thickness can be minimized by

Small lens size

High index materials

Aspheric lens

Small lens size

- Lens edge gets thicker farther away from the center
- Frames with rounded corners should be used



- A frame wider than wearer's face at the temple area avoided



High minus lens makes the side of wearer's head look narrower through the lens

- Excessive decentration avoided- outer lens edge will be much thicker than the inner edge



alternative

Use a wider bridge and smaller eye size

High index materials

Normal index	$n \geq 1.48$ but < 1.54
Mid index	$n \geq 1.54$ but < 1.64
High index	$n \geq 1.64$ but < 1.74
Very high index	$n \geq 1.74$

BS 7394: Part 2, "*Specification for complete spectacles*"

Advantages

1) Thinner

2) Lighter

refractive index ↑ density ↑

✓ to compare the weight of lenses made of diff materials , consider the saving in volume

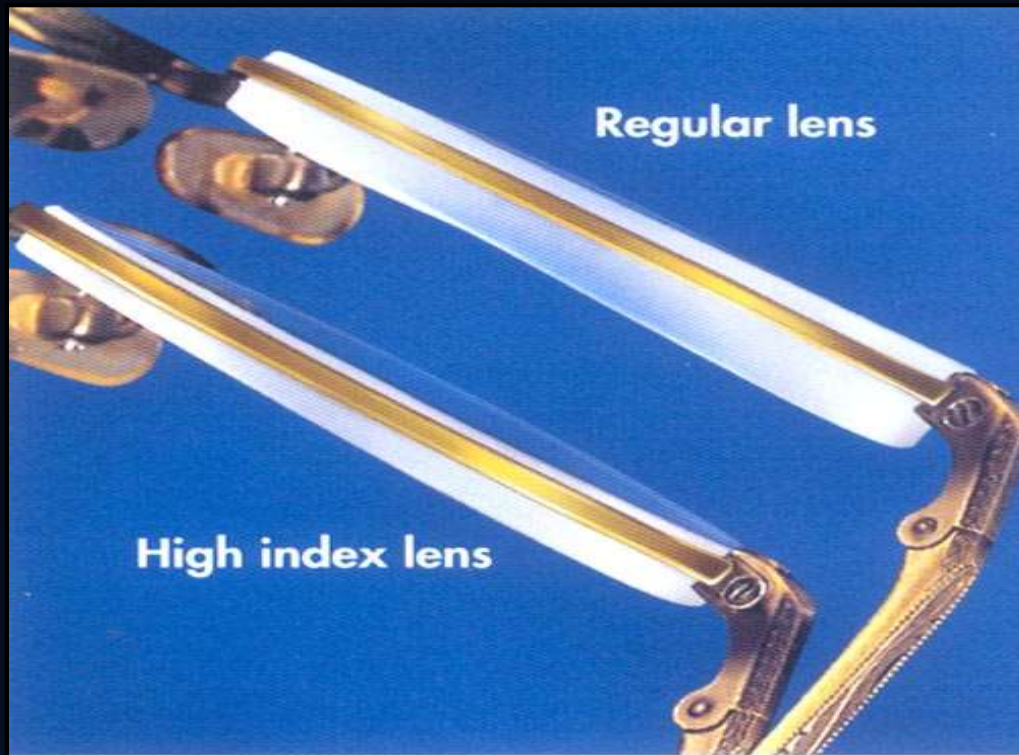
✓ Saving in weight due to increase in density, for a given power

Weight reduced in high powers

Weight increased in low powers

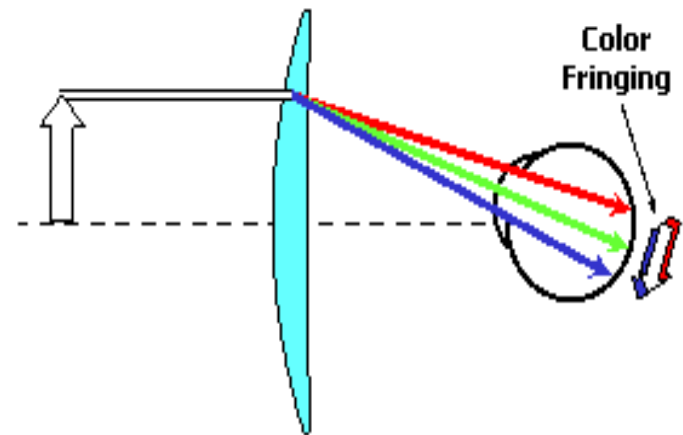
3) Good cosmesis

4) Magnification reduced



Disadvantages

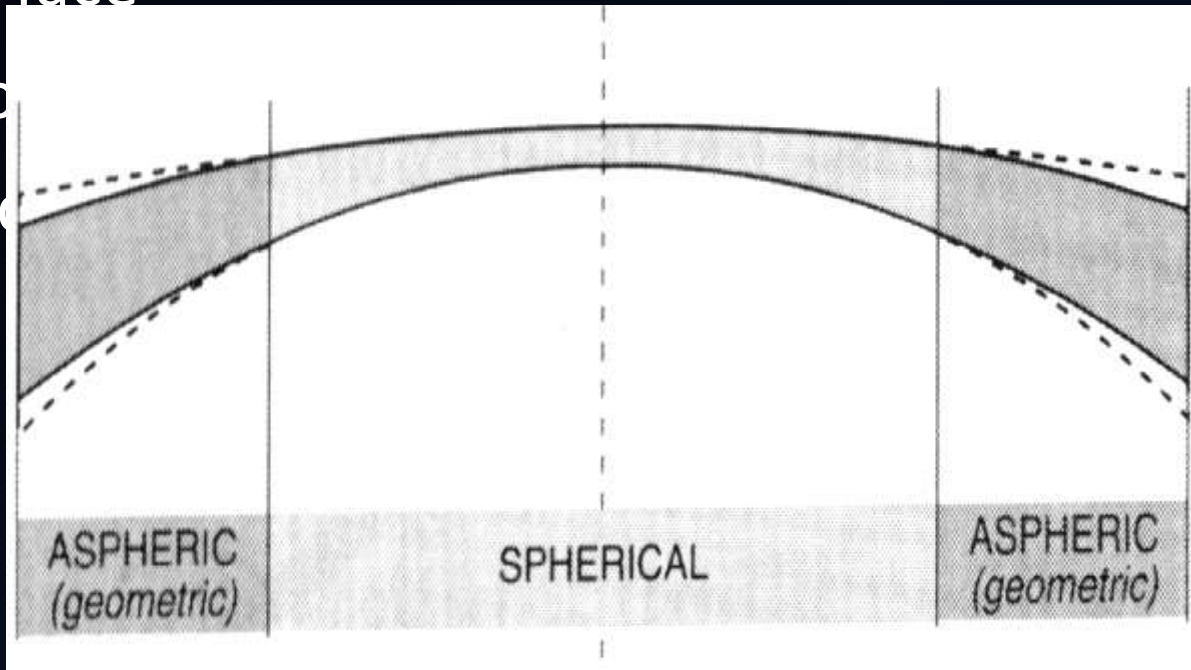
- Higher the R.I , more is the reflection, thus reduced transmission
- Have lower Abbe value, hence increasing TCA



"Color Fringing" of Lateral Chromatic Aberration

Aspheric lens

- A lens in which one or both surfaces are not spherical
- Doesn't have same radius of curvature over the entire surface



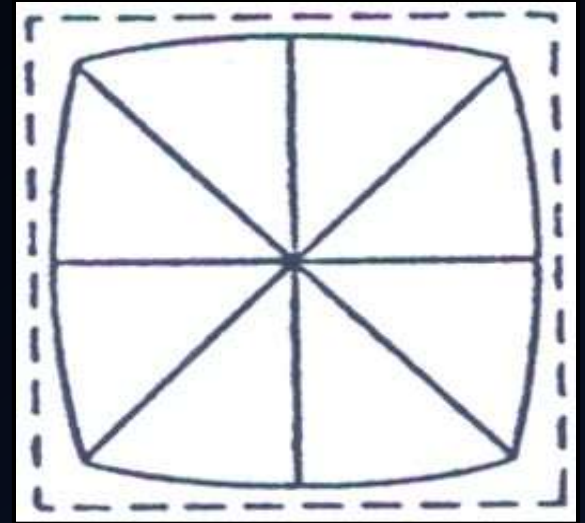
- Optically correct oblique astigmatism produced when looking obliquely through lens
- In middle ,starts out as spherical surface
- At a certain distance from OC , lens changes its curvature at a rate calculated to offset peripheral aberration.
- Curvature changes both radially and circumferentially

- 
- Power change along the radial direction is 3 times more than change in circumferential direction
 - This inherent astigmatism counterbalance oblique astigmatism

- 
- **Edge thickness can be masked**
 - By using dark plastic frames with the beveled edge of the lens colored to match the frame
 - The edge bevel can be positioned forward in such a way that frame hides the bulk of the thick edge

2. Minification

- High minus lenses



Cause minification of wearer's eyes to the observer

Minify the retinal image size

- Both the apparent size of the wearer's eyes and the retinal image size increased by



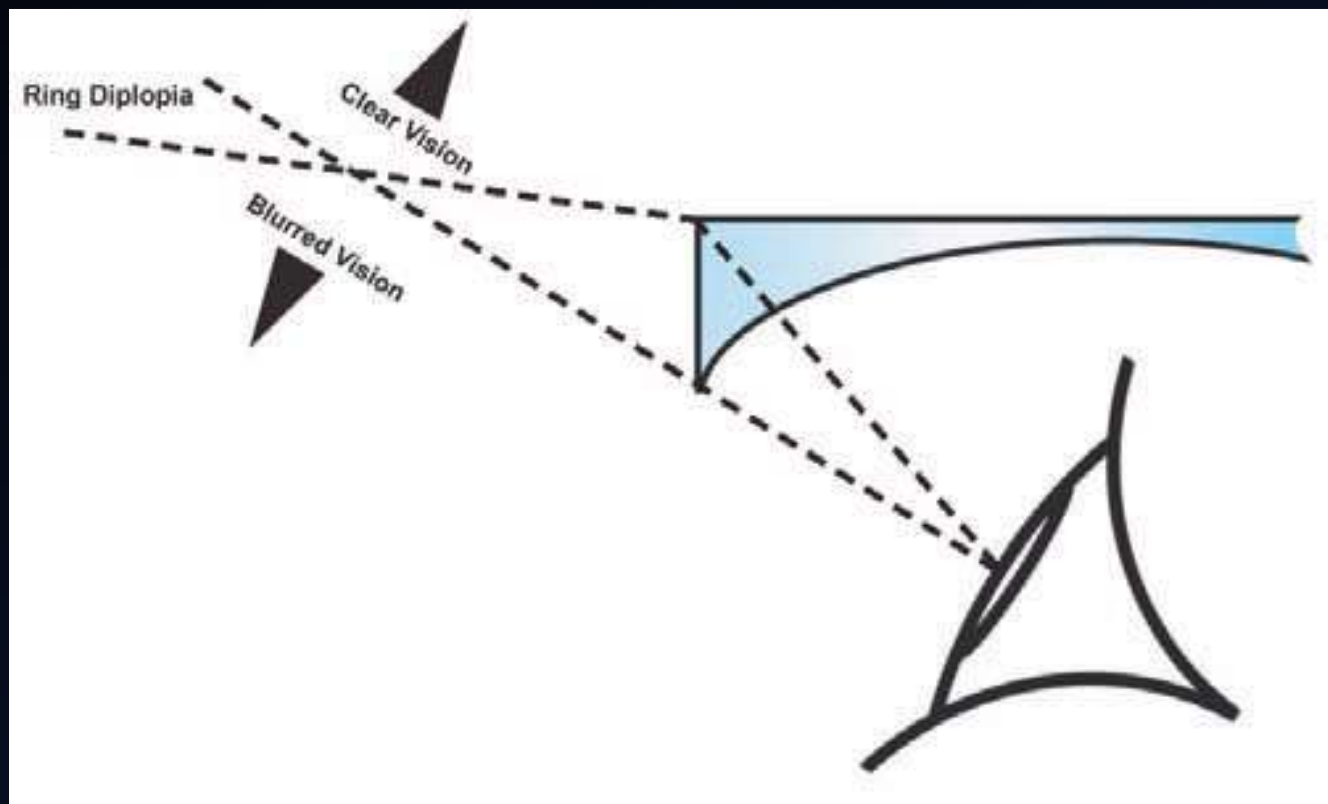
Fitting the lenses closer to wearer's eye

Contact lens

3. Field of View

- Minus lenses increase both the peripheral and macular fields of view
- Minus lenses produce an area of overlap at and around the lens margin
- Myope may experience a ring shaped area of overlapping double vision- an area that involves a relatively clear vision through the lens but blurred vision outside the lens

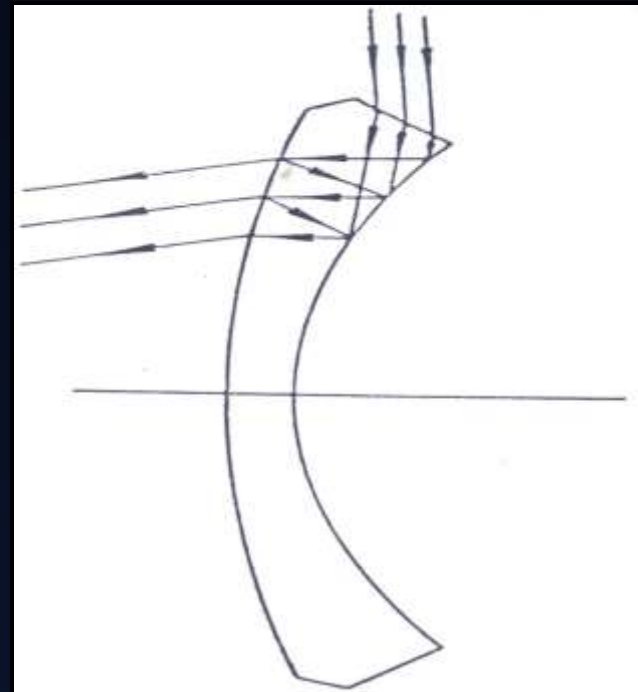
Ring diplopia is so far in the periphery that it is seldom noticed



4. Edge reflection & concentric rings

- In addition to the reflections formed by the surfaces, edges of the high minus lenses produce multiple rings seen just inside their edges
- These are true images of the edge of lens
- Caused by total internal reflections that begin at the lens edge and reflect their way toward the center of lens until they strike the surface at an angle that permits refraction by lens surface and into the eye of observer

- Process is then repeated and multiple ring reflections seen by observer
- Rings more pronounced on oblique viewing than straight ahead position



- Antireflection coating attenuate the intensity of rings on straight ahead viewing
- For rings seen on oblique viewing, effective method is edge treatment
 - a. Edge coating – reduce the granular appearance of the edge
 - b. Edge painting – painting the edge with neutral gray color make rings less noticeable
 - c. Semitransparent edge – for a glass lens produced by using an edging wheel

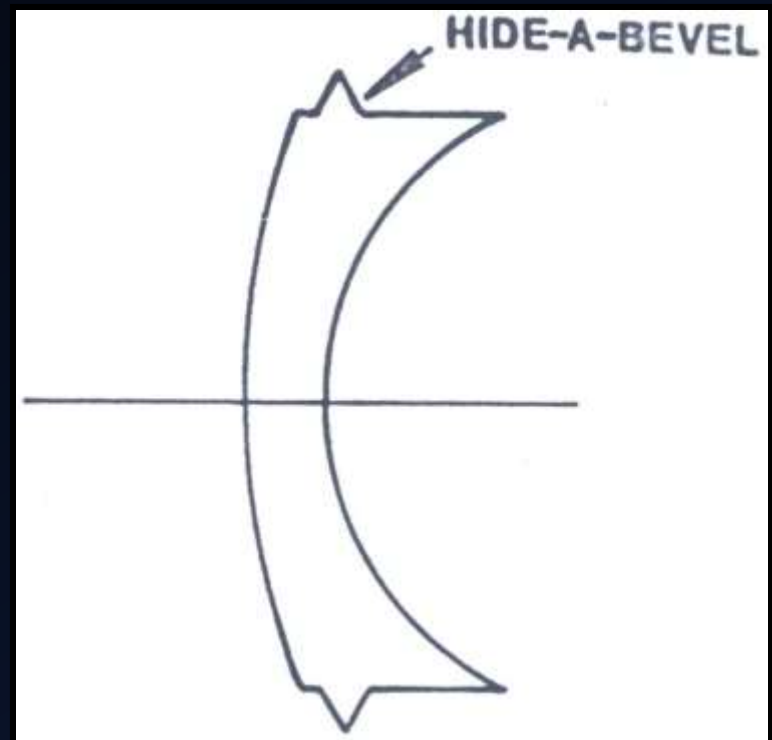
d) Buffing the edge – for a plastic lens, produce same effect as a semitransparent edge on glass lens

e) Tinting the lens – tinting the lens after it is edged

f) Hide-a-Bevel technique

- Consists of a flat edge with a narrow bevel protruding from flat edge
- With such edge, there is a smaller area of ground glass from which reflections can arise

- The angle made by flat edge with lens surface is such that the rays tend to be reflected so that they do not enter the observer's eye



Lenticular lenses

- Used in cases of very high myopia, in excess of 11.00 or 12.00 D
- The central area of the lens contains the prescribed refractive power of the lens
- The peripheral area i.e the carrier serves only to extend the physical size of lens without increasing its thickness



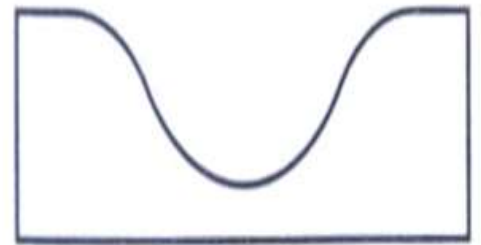
Forms



Myodisc



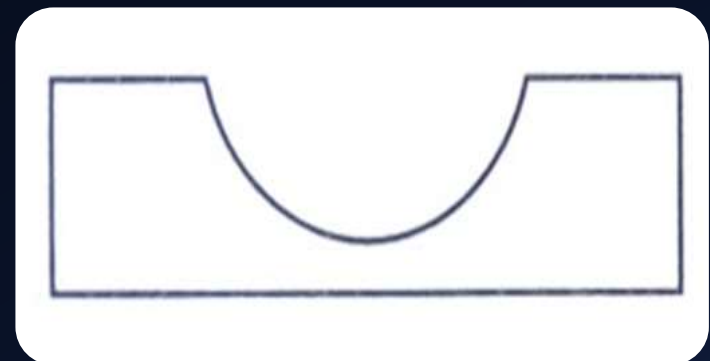
**Minus
lenticulars**



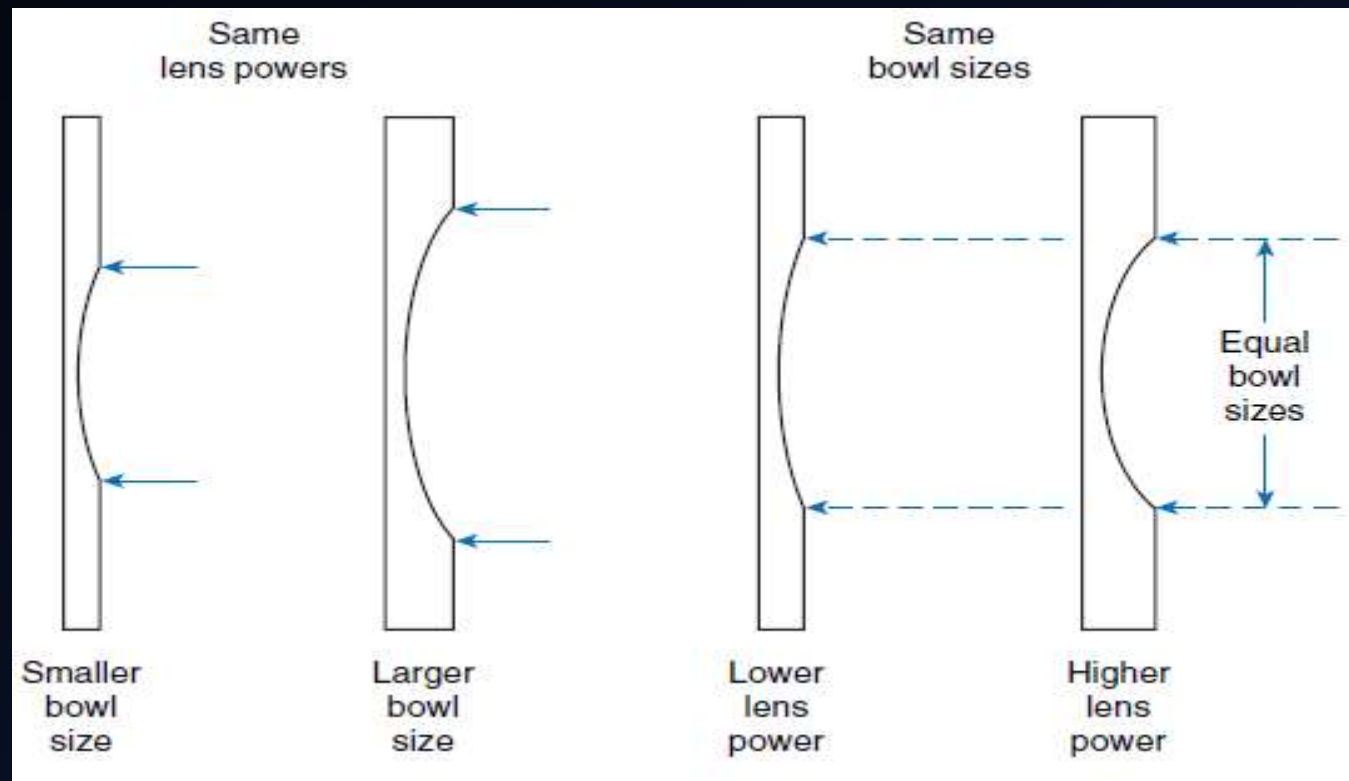
**Blended
myodisc**

Myodisc

- Front surface is either flat or almost flat
- Front surface usually contains the cylinder component of prescription
- A high minus “bowl” in the middle of back surface
- Plano back carrier area

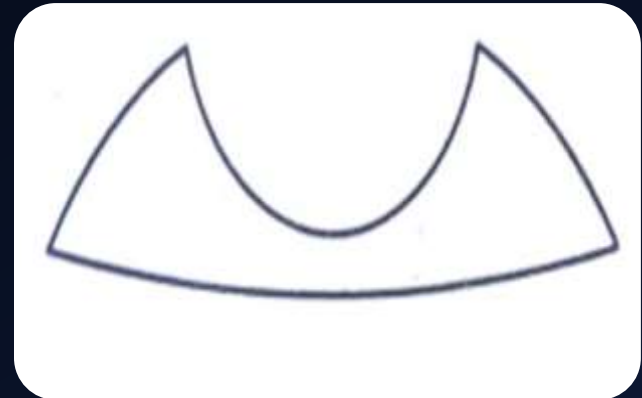


- Larger the bowl area , thicker will be the carrier
- For lenses with same sized bowl areas, increase in lens power will mean an increase in carrier thickness



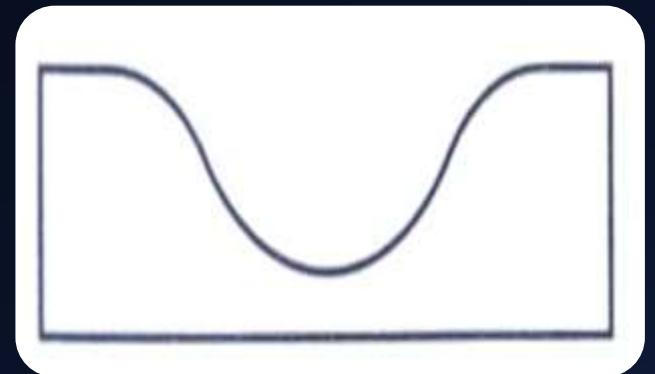
Minus lenticular

- Carrier is not plano
- Instead back surface of the carrier is made positive so that the outer edge will thin down considerably



Blended lenticulars

Optical aperture is blended into the flange eliminating the obvious line and improving the appearance



Disadvantages of lenticulars

- Bull's eye appearance (fried egg)
- Expensive



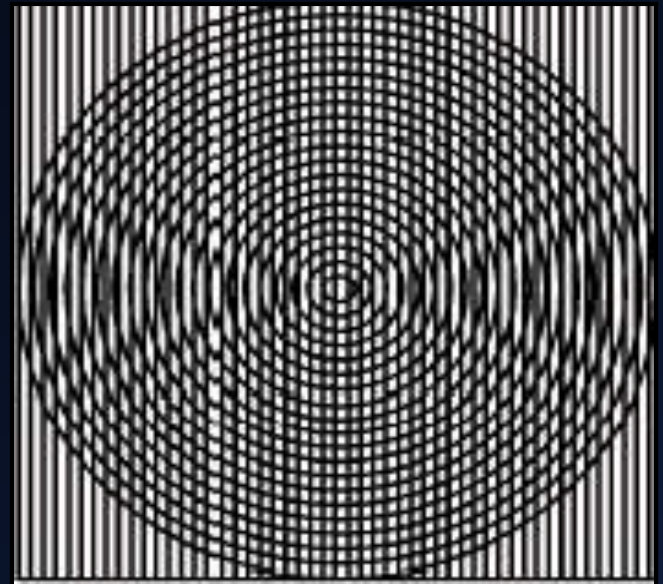
Fresnel Press on Lenses

- Made in same manner as Fresnel press on prism
- Grooves in the material form a concentric pattern on the surface instead of parallel lines
- Successive concentric grooves cut away from the center have slightly increased apical angles



Increased deviating power from center to periphery

- Available from -1.00 to -14.00 D
- Cylinders power not available
- Slight reduction in visual acuity
- Reduced contrast
- Visibility of concentric grooves to the observer



Use

Smaller eye size

If in between sizes, use the smaller indicated eye size and larger indicated bridge size

Rounded corners

Low density (light weight) lenses if weight reduction is important

Midindex or high index lenses, such as polycarbonate or high index plastic

Flatter or "hidden" bevels

Polished or rolled and polished edges

Without rolled and/or polished edges, use frames with rims that hide lens edges combined with bevel placement that balances

Antireflection coating or a light tint (AR coating is preferred over a light tint)

Aspheric or atoric lenses

For exceptionally high minus—biconcave or minus lenticular lenses

Avoid

Large lenses

Excessive decentration

Squared-off corners

Crown glass lenses

Low index lenses, such as CR-39 plastic lenses

Full V bevels

Nonpolished, frosted-looking edges causing a concentric-ring appearance

Thin rims in combination with lenses having unpolished edges

Frames that extend beyond the side of the wearer's head

Flat, non-AR-coated front curves

CONTACT LENSES

- Lesser image minification
- Effectivity relationship:

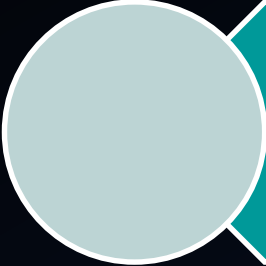
$$F_c = F_s / 1 - dF_s$$

power of lens is reduced

- Wider field of view
- Minimal aberrations
- Lesser prismatic effect
- Better cosmesis

- 
- Greater accommodative demand
 - More amount of convergence needed

REFRACTIVE SURGERY



Radial keratotomy



Photorefractive
keratectomy



LASEK and Epi-LASIK



LASIK



Clear lens extraction



Phakic intraocular lens

Radial keratotomy

- Involves use of radial incisions
- Arranged in concentric pattern surrounding a central clear zone
- Results in central flattening and peripheral steepening of cornea
- Limited to about -3.00 D correction
- Obsolete nowadays

Photorefractive keratectomy

- Made the first use of excimer laser in 1980
- Excimer laser : mixture of
 - argon 0.5% to 12%
 - fluoride 0.5%
 - remainder – buffer rare gas like helium or neon
- Emits laser radiation at wavelength of 193 nm
- Provides photoablation with minimal collateral damage

Steps

- Separation or removal of corneal epithelium using rotating brush
- Bowman's m/m exposed
- Corneal collagen lamellae, keratocytes and GAGs are targets of photoablation
- Epithelium grows back within one week
- Visual recovery occurs as epithelium becomes more regular

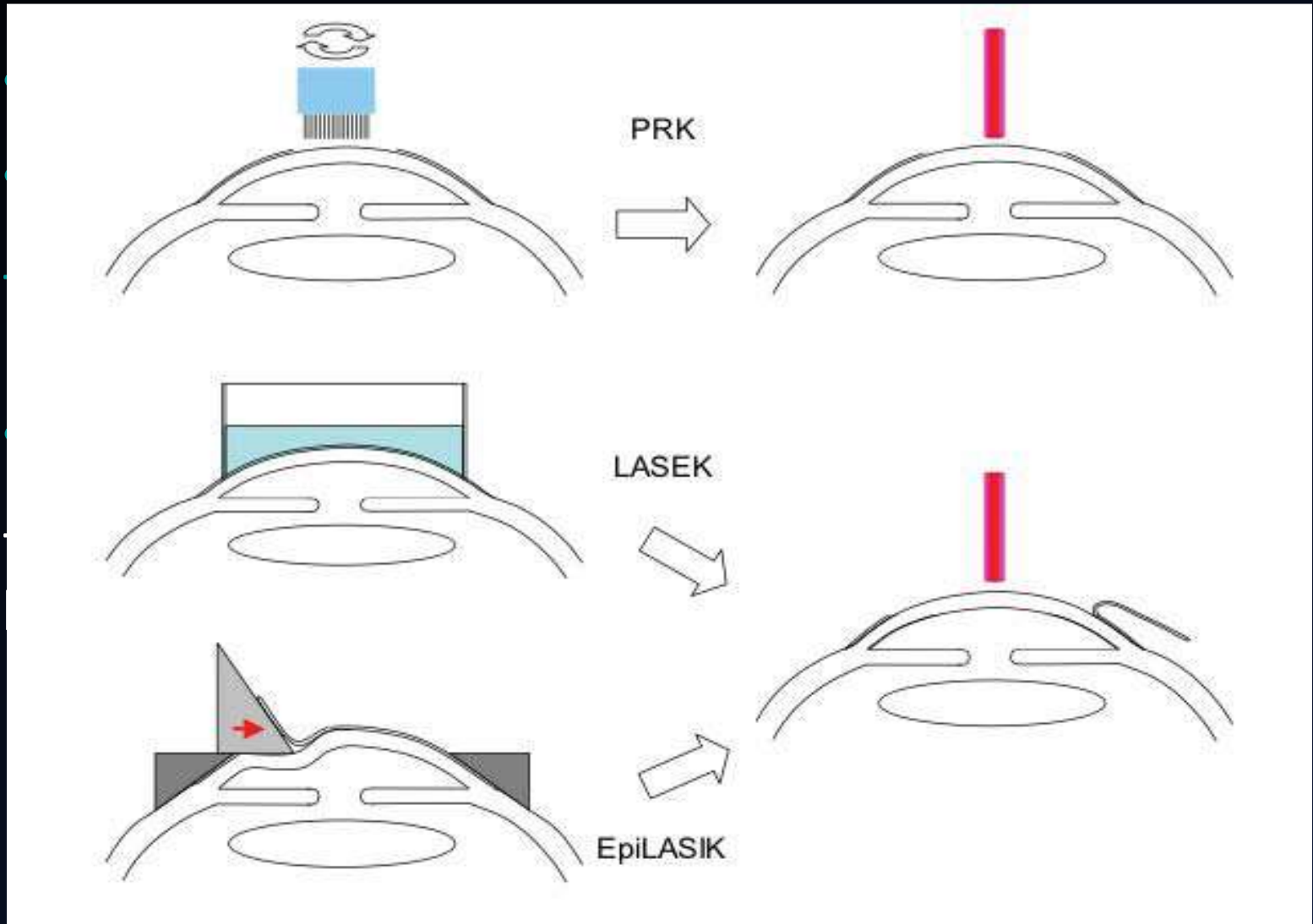
Side effects

Post operative pain

Slow visual recovery

Corneal haze

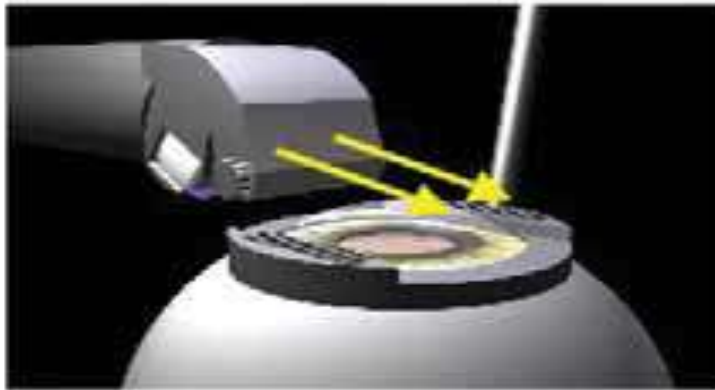
LASEK and Epi- LASIK



LASIK (laser assisted in-situ keratomileusis)

- Introduced in 1990
- Keratomileusis Gr word : keras (horn like or cornea) smileusis (carving)
- Minimize the problems of PRK
- Combines two techniques of surgery to correct refractive error
 - 1) A laser called femtosecond laser (intraLASIK)
 - 2) Complex automated surgical blade called microkeratotome (LASIK)

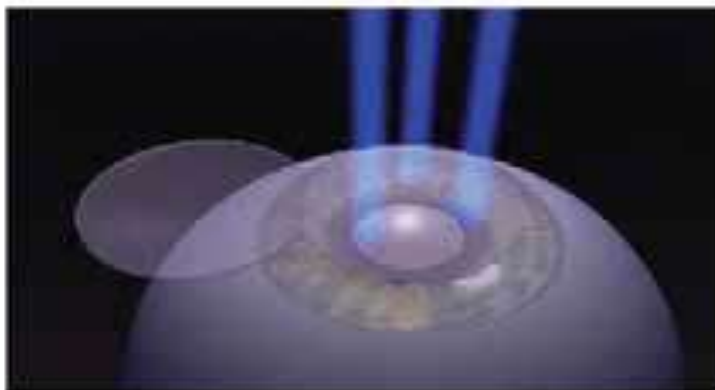
to create a thin flap in cornea



1 Flap creation



2 Flap lift



3 Excimer laser reshaping



4 Flap is reflected back

Complications of LASIK



Undercorrection



Overcorrection



Glare and halo



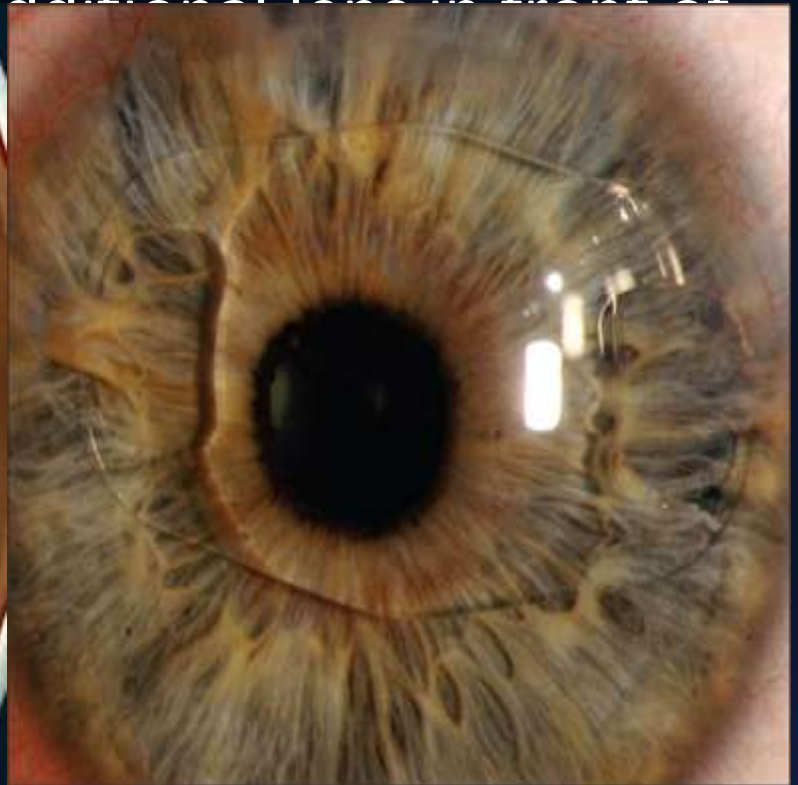
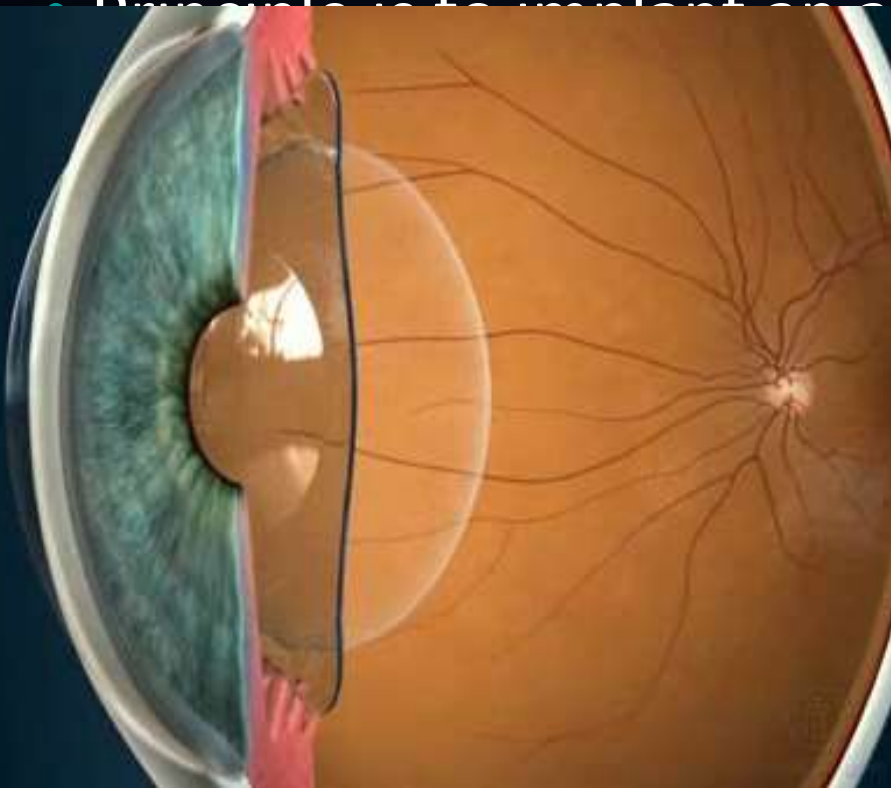
Dislodged flap and flap wrinkle



Dry eye syndrome

Phakic Intraocular Lens

- Indicated in higher errors > -10.00 D or patient with thin or abnormal corneas
- Principle is to implant an additional lens in front of



Clear Lens Extraction

- Removal of non cataractous crystalline lens
- Effective in cases of severe myopia usually > -10.00 D
- Increased risk of retinal detachments
- Loss of accommodation

Laser refractive surgery

(LASIK and surface surgery)

Myopia -1.00 to -10.00 D

Hyperopia +1.00 to +4.00 D

Astigmatism 1.00 to 6.00 D

Intraocular refractive surgery

(Phakic IOL if < 45 or RLE if > 45 years old)

Myopia > -10.00 D

Hyperopia > +4.00 D

Astigmatism 1.00 to 6.00 D

General Criteria To Be Met For Refractive Surgery

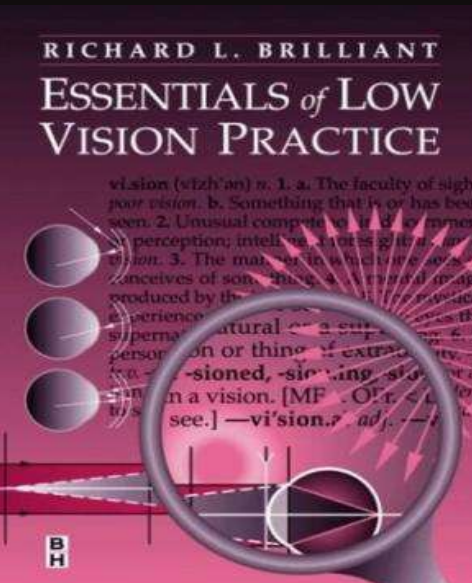
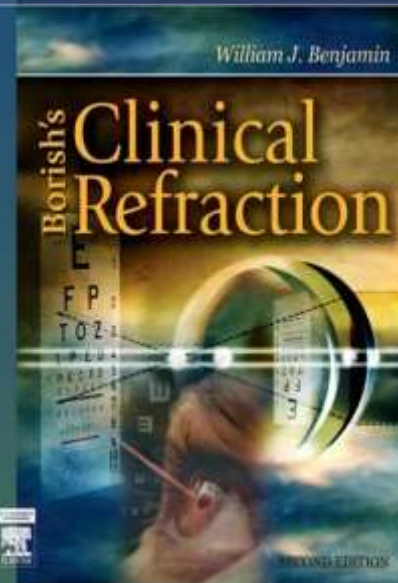
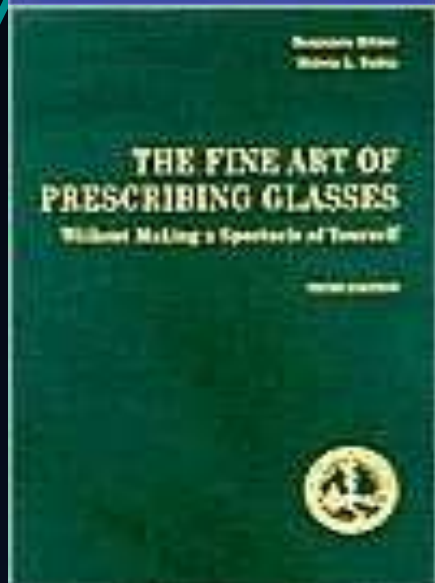
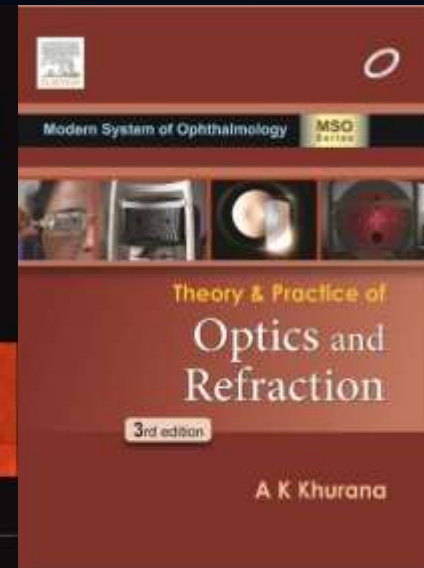
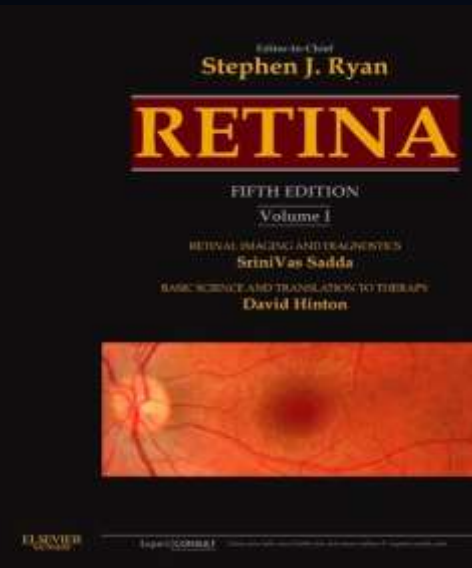
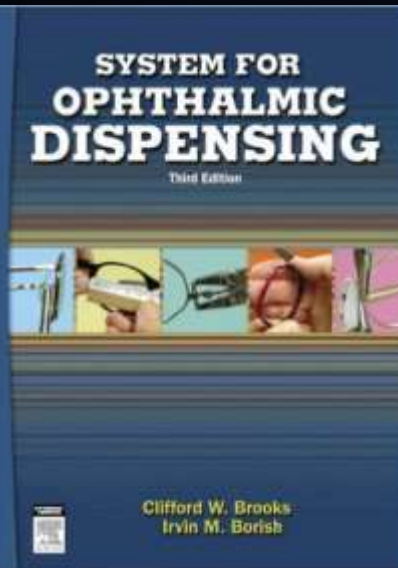
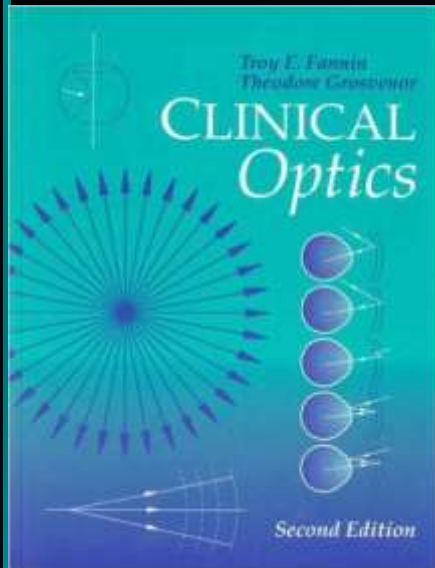
- Desire to be more independent of spectacles / contact lenses
- Stable refractive error (less than 0.50 DS change over the last two years)
- Age greater than 18
- Realistic expectations and understanding of risks related to refractive surgery
- Not pregnant or breastfeeding
- No significant ocular pathology or general health problems

Low vision management of high myopia

- Pathological myopia is one of the leading cause of visual impairment and blindness worldwide
- For magnification at near
 - Remove spectacle correction and use the myopic system as a built in microscope at the appropriate working distance
 - Contact lens wearers can wear microscopic spectacles over contact lens
- Other magnification devices for distance and near are effective as well

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- Sunlenses to eliminate photophobia outdoors
 - If night vision is poor , a flashlight may be helpful

REFERENCES



Thank You

